

Phase 1 – Reconnaissance (Information Gathering)

Objective:

The objective of the reconnaissance phase is to gather basic information about the target machine before performing any security assessment. This phase helps determine whether the target is reachable and prepares the tester for further scanning activities.

Purpose:

Reconnaissance is the first phase of penetration testing. During this phase, no attempt is made to exploit the target system. Instead, information is collected to understand the target's network presence and accessibility. This information is essential for planning the subsequent stages of the penetration test.

Step 1: Identify the Target Machine:

The target machine used for this assessment was **Metasploitable2**, and the attacking machine was **Kali Linux**. Both virtual machines were connected using a Host-Only network in VirtualBox.

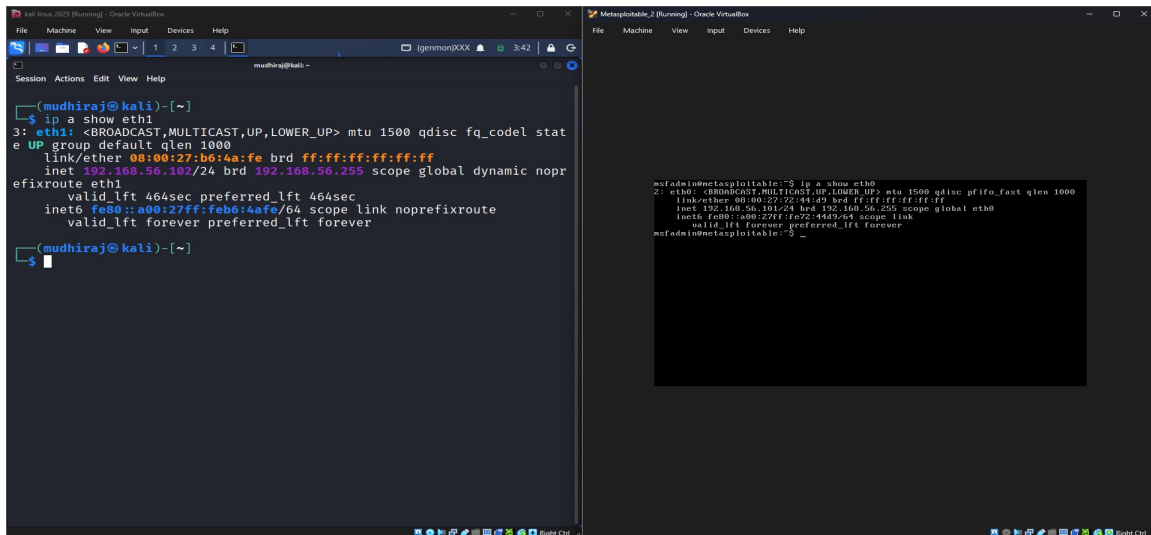


Figure 1: IP configuration of Kali Linux (192.168.56.102) and Metasploitable2 (192.168.56.101).

Attacker Machine (Kali Linux): 192.168.56.102

Target Machine (Metasploitable2): 192.168.56.101

Step 2: Verify Network Connectivity:

Before starting any scan, network connectivity between Kali Linux and the target machine was verified using the ping command.

Command Used

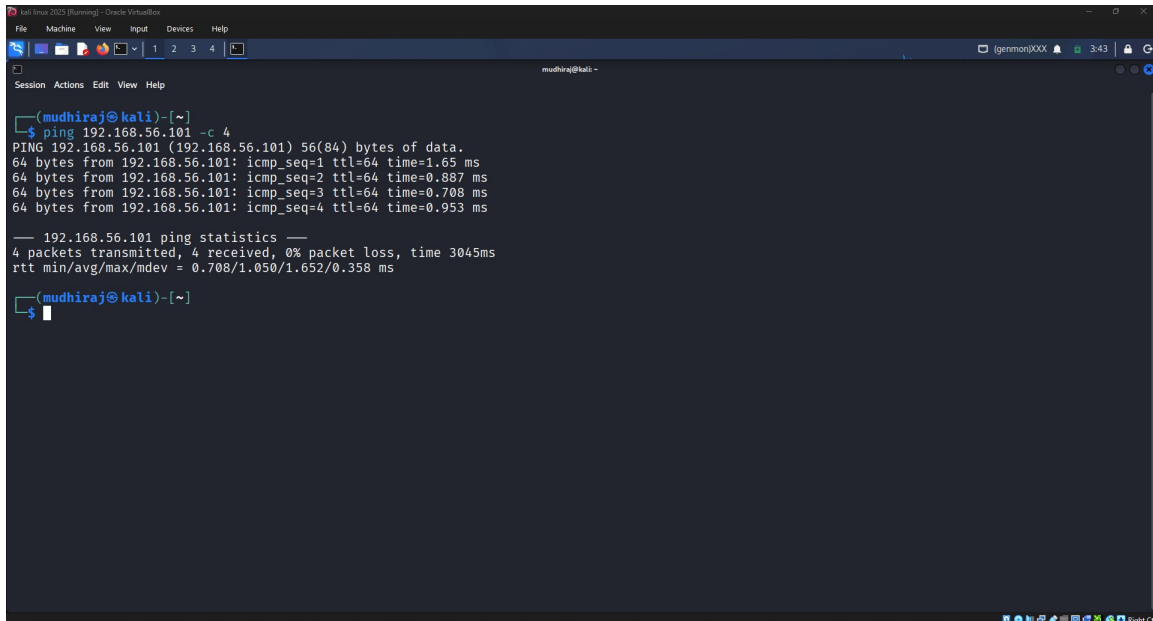
```
ping 192.168.56.101
```

Command Explanation

- ping sends ICMP Echo Request packets to the target.
- If the target replies, it confirms that the machine is online and reachable through the network.
- It also provides information about packet loss and response time.

Observation:

The target system responded successfully to all ICMP Echo Requests with **0% packet loss**, confirming that the machine was powered on, reachable, and ready for further assessment.

A screenshot of a Kali Linux terminal window. The terminal shows the execution of the command 'ping 192.168.56.101 -c 4'. The output displays four successful ping responses with varying response times (1.65 ms, 0.887 ms, 0.708 ms, and 0.953 ms) and a summary indicating 0% packet loss and a total time of 3045ms. The terminal window has a dark background with light-colored text. The top of the window shows the 'Kali Linux 2021 (running)' title bar and standard window controls. The bottom of the window shows the system tray with various icons and the 'Right Click' text.

```
(mudhiraj@kali)-[~]
$ ping 192.168.56.101 -c 4
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data:
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=1.65 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=0.887 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=0.708 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=0.953 ms

--- 192.168.56.101 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3045ms
rtt min/avg/max/mdev = 0.708/1.050/1.652/0.358 ms
(mudhiraj@kali)-[~]
$
```

Figure 2: Successful ICMP connectivity verification using the ping command with 0% packet loss.

Reconnaissance Summary:

The reconnaissance phase confirmed that the target machine was accessible over the network. The target IP address was successfully identified, and communication between Kali Linux and the target machine was verified. Since the target responded correctly to the connectivity test, the assessment proceeded to the Scanning phase using Nmap.